

Learning Intention: I can use mental strategies to add and subtract numbers efficiently.

Success Criteria — I can:

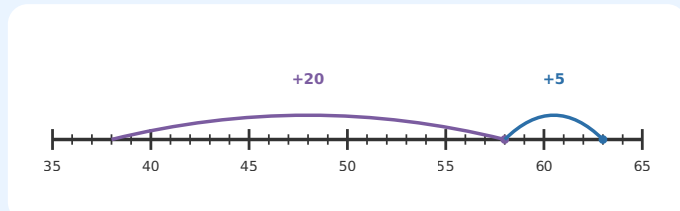
- Use doubles and near-doubles to solve addition facts quickly.
- Use a jump strategy on a number line to add or subtract.
- Use the split (partitioning) strategy to add two-digit numbers.

Strategy Box — Three Mental Strategies

Strategy	How it works
Near doubles	Use a known double, then adjust by 1 or 2 (e.g. $7+8 = 7+7+1$)
Jump strategy	Start at one number and jump forward/back in friendly chunks (tens, then ones)
Split strategy	Break numbers into tens and ones, add each part, then combine

Worked Example — $38 + 25$

Jump strategy (number line):



Start at 38. Jump +20 to 58. Jump +5 to 63.

Both strategies give the same correct answer — 63. Choose whichever feels easiest for the numbers in front of you.

Split strategy:

$38 = 30 + 8$
 $25 = 20 + 5$
 Tens: $30 + 20 = 50$
 Ones: $8 + 5 = 13$
 Combine: $50 + 13 = 63$

Review: Near Doubles

If you know $7 + 7 = 14$, then $7 + 8$ is just **one more**: $14 + 1 = 15$.
 If you know $8 + 8 = 16$, then $8 + 7$ is **one less**: $16 - 1 = 15$.

Try It

Use near doubles to solve: $6 + 7 = \underline{\quad}$ $9 + 8 = \underline{\quad}$

Parent Tip

Practise mental strategies during everyday moments — adding up shopping items, working out time until an event, or splitting a group of lollies. Ask "how did you work that out in your head?" to hear their strategy.

Key Vocabulary

Word	Meaning
Partition	To split a number into parts (e.g. tens and ones)
Efficient	Solving a problem in a quick, sensible way
Mental strategy	A method for solving a problem in your head, without writing every step

Build Skills — Apply mental strategies to solve problems.

Q1 — Near Doubles

Use a near-doubles strategy to solve each fact. Show the double fact you used.

Fact	Double fact used	Answer
$6 + 7$		
$9 + 8$		
$5 + 6$		

Q2 — Jump Strategy on a Number Line

Solve $46 + 27$ using the jump strategy. Draw your jumps on the number line below.



Answer: $46 + 27 =$ _____

Q3 — Split Strategy

Use the split strategy to solve each problem. Show your working.

Problem	Tens + Tens	Ones + Ones	Answer
$34 + 42$			
$57 + 26$			

Q4 — Choose Your Strategy

Solve $64 - 28$ using any mental strategy from this worksheet. Show your working and name your strategy.

Watch Out!

When using the split strategy, remember to keep tens with tens and ones with ones — mixing them up (e.g. adding a ten to the ones) is a common error. Always check your combined answer makes sense.

Apply & Reasoning — Use what you know to explain your thinking.

- 1 Reasoning:** Explain why knowing $7+7=14$ helps you solve $7+8$ quickly. What do you do to the answer?

- 2 Reasoning:** For $52 + 39$, would you use the jump strategy or the split strategy? Explain which is more efficient for you and why.

- 3 Open-ended:** Solve $47 + 36$ using **two different strategies** from this worksheet. Show both, and check they give the same answer.

Quick Check — Exit Ticket

Solve $8 + 9$ using a near-doubles strategy. Show which double fact you used.

Answer & Teacher Support**Answer Key — Pages 1 & 2**

Try It (p1): $6+7=13$ (using $6+6=12$, $+1$); $9+8=17$ (using $8+8=16$, $+1$, or $9+9=18$, -1).

Q1 (p2): $6+7=13$ ($6+6=12+1$); $9+8=17$ ($9+9=18-1$ or $8+8=16+1$); $5+6=11$ ($5+5=10+1$).

Q2 (p2): $46+27=73$ (e.g. $46+20=66$, $+7=73$).

Q3 (p2): $34+42$: $30+40=70$, $4+2=6$, total 76. $57+26$: $50+20=70$, $7+6=13$, total 83.

Q4 (p2): $64-28=36$. Accept any correct strategy (e.g. jump: $64-20=44$, $-8=36$; or $64-30=34$, $+2=36$).

Answer Key — Page 3

Q1: $7+8$ is one more than $7+7$, so add 1 to 14 to get 15.

Q2 (acceptable-answer guide): Accept either strategy with a valid efficiency reason (e.g. jump is quicker because 39 is close to 40). Correct answer either way: 91.

Q3 (acceptable-answer guide): Accept any two correct, genuinely different strategies (e.g. split and jump) that both correctly reach 83.

Exit ticket: $8+9=17$, using $8+8=16+1$ or $9+9=18-1$.

Teacher Checklist

- ☐ Uses near doubles to solve addition facts to 20.
- ☐ Applies the jump strategy accurately on a number line.
- ☐ Applies the split strategy, keeping tens and ones separate.
- ☐ Selects and justifies an efficient strategy for a given problem.
- ☐ Solves the same problem with two different strategies.

Teacher Tip

Watch for students defaulting to written algorithms instead of mental strategies. Encourage them to estimate first, then choose a strategy that matches the numbers (e.g. jump strategy works well when one number is close to a multiple of 10).